

ZACHARY LUKAN

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EDUCATION

McGill University, Montreal

Graduating class of 2029

Major Mathematics and Computer Science (B. Sc.)

Related Coursework: Data Structures & Algorithms, Object-Oriented Programming, Multivariate Calculus, Programming Paradigms, Computer Architecture, Statistics & Probability, Discrete Math, Algorithmic Problem Solving

PROJECTS

Facial Tracking & Emotion Recognition Using Neural Networks

OpenCV, PyTorch, FastAPI, NiceGUI, Ax

- Engineered a real-time facial emotion recognition system using OpenCV-based face tracking and a PyTorch CNN, transforming live video streams into emotion classifications with 77% accuracy and 15 fps
- Developed and deployed a production-ready ML application stack with FastAPI, Docker, and NiceGUI, exposing model inference through an interactive web interface with cross-platform accessibility
- Increased model performance through Bayesian hyperparameter optimization using Ax, automating experimentation across training configurations and accelerating convergence resulting in a 80% increase in performance

Panorama Stitcher with Blending

OpenCV, Numpy, Streamlit

- Built an image-stitching pipeline achieving a Blind Image Spatial Quality score within 5% of state-of-the-art applications
- Implemented RANSAC-based homography estimation for perspective alignment robust to outlier feature matches
- Integrated a multi-band Laplacian pyramid blending algorithm to eliminate seams, boosting accuracy by 56%
- Built an interactive Streamlit frontend for uploading, ordering, and stitching photos in-browser

LEADERSHIP EXPERIENCE

Canadian CanSat Launch Campaign

June 2025

Lead of Software Development

Lethbridge, AB

- Co-designed and manufactured a compact FPV drone that withstands 18 G of force and self-stabilizes, streaming to a VR headset to earn the technical design award
- Programmed embedded C++ firmware that streams air pressure, position, and force data to the ground station in real time over a 15km+ range
- Presented results to a panel of judges and reached over 1 million viewers online

University of British Columbia Physics Olympics

March 2025

Team Lead

Vancouver, BC

- Led a team of developers and engineers to victory against 80 teams in a technical design and laboratory competition
- Directed teams of 7-10 members, running team meetings and overseeing multiple concurrent engineering projects

TECHNICAL STRENGTHS

Computer Languages

Java, Python, C/C++, HTML/CSS, OCaml

Frameworks

OpenCV, PyTorch, NumPy, Ax, FastAPI

Technical

Data Structures & Algorithms, Computer Vision, Neural Networks, Statistical Methods

Tools

Git/GitHub, Google Cloud, Microsoft Office, Google Colab, Docker